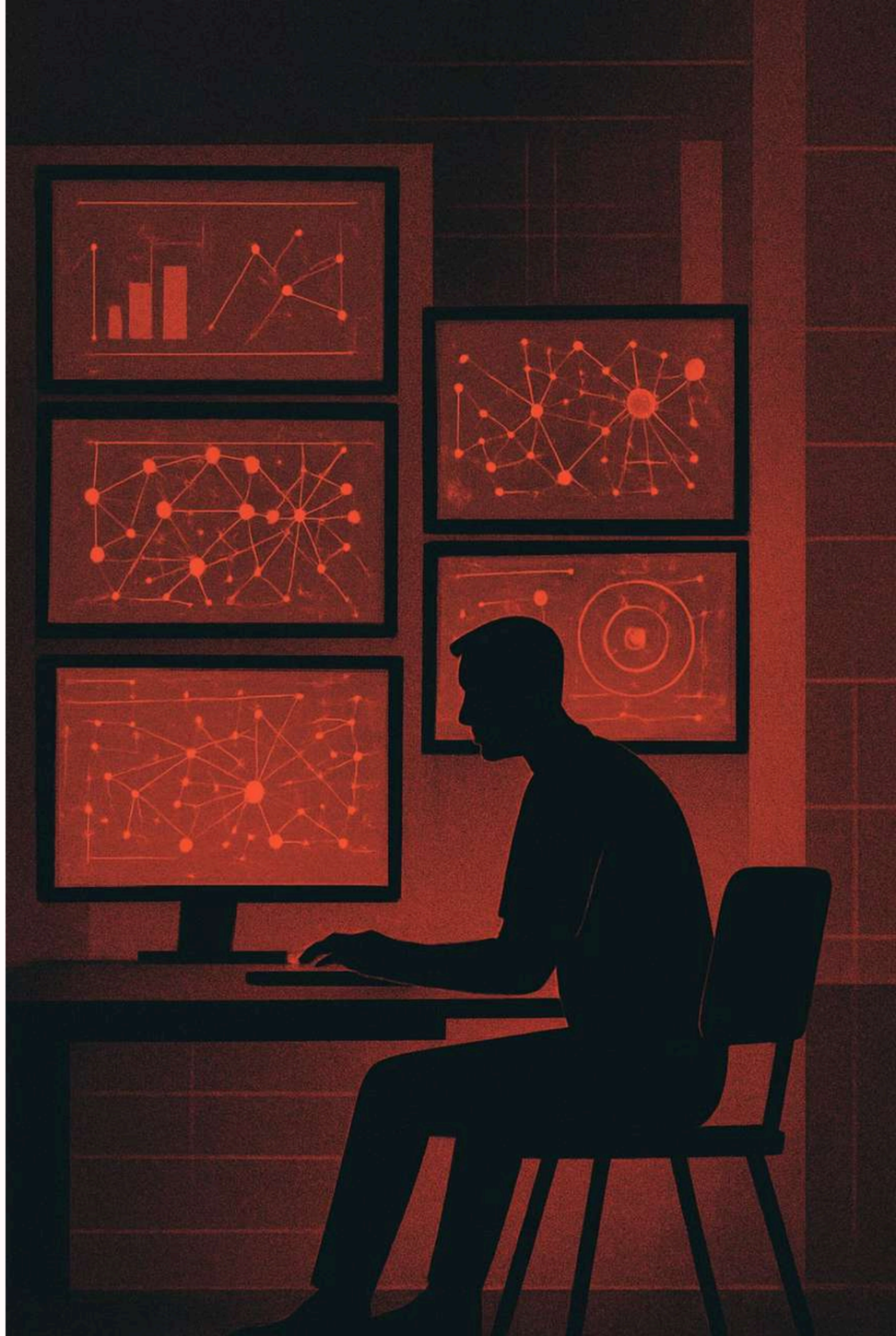
Customer Churn Analysis & Retention Strategy

A comprehensive analysis of telecom customer churn using Python, SQL, and Power BI — identifying key drivers and data-driven strategies to improve retention.



Dataset at a Glance

7,043

Total Customers

21

Columns

1,869

Churned

5,174

Retained

Key Numerical Features

- **Tenure:** 0–72 months (avg. 32.4)
- **Monthly Charges:** $18.25–$118.75 (avg. $64.76)
- **Total Charges:** up to $8,468
- **Senior Citizens:** 16.2% of customers

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Data Preparation with Python

Data Loading

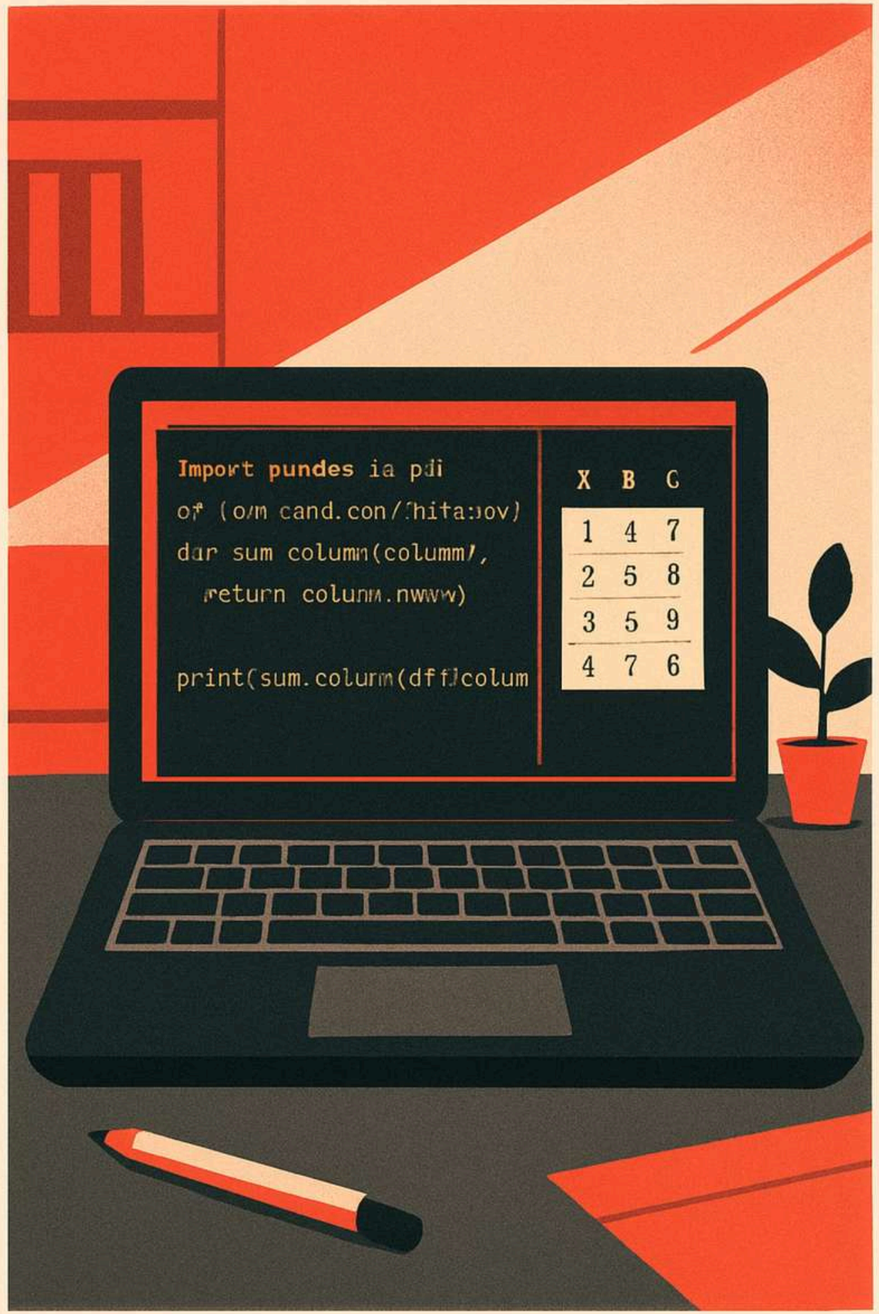
Imported dataset via pandas; used df.info() and df.describe() for structure and summary statistics.

Missing Values

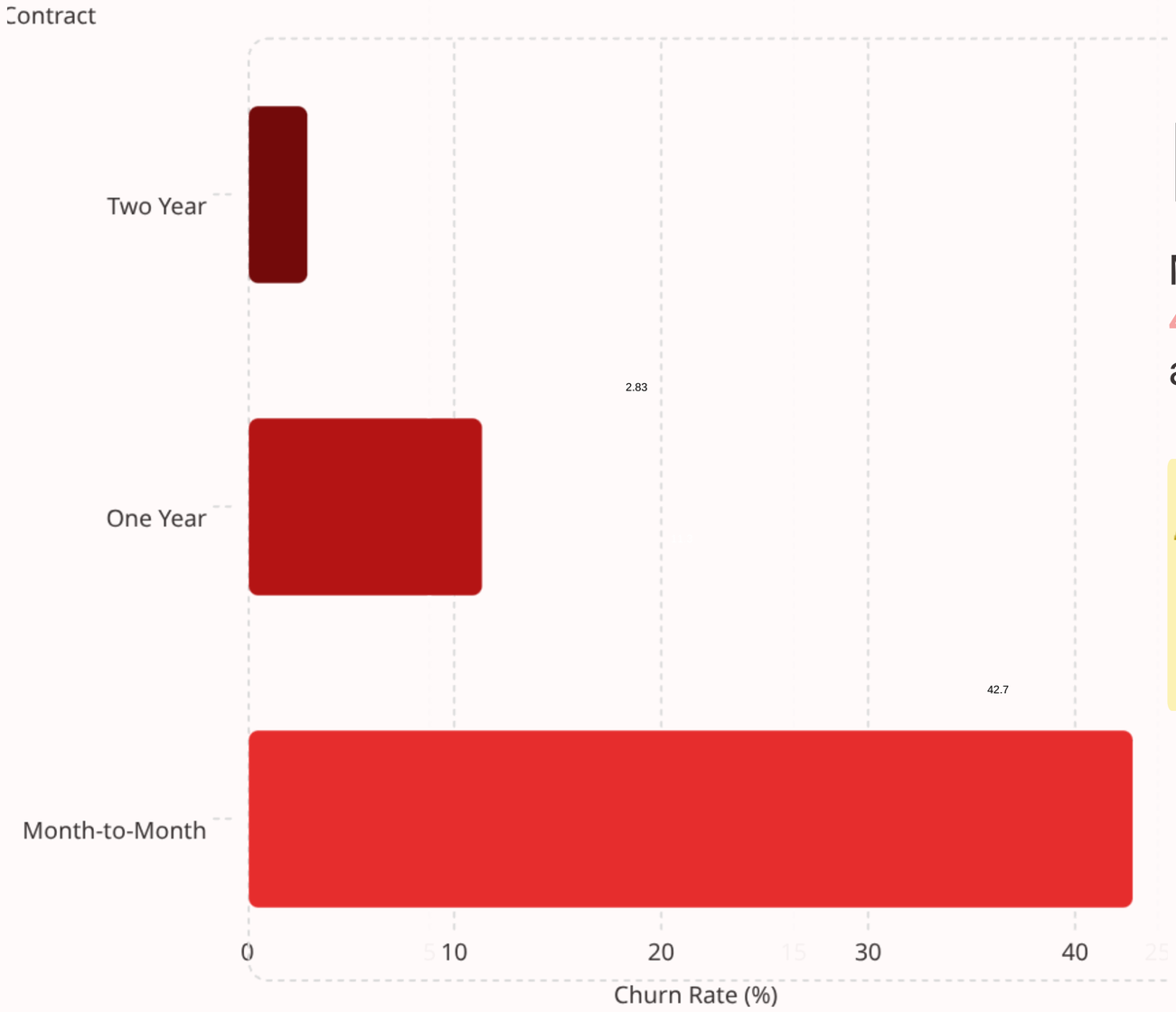
15 nulls in PhoneService, **115** in PaperlessBilling — filled using column mode values.

Duplicates

Zero duplicate rows confirmed via dp.duplicated().sum().



Churn by Contract Type



Key Finding

Month-to-month customers churn at a staggering 42.71% — nearly 4× the rate of one-year contracts and 15× that of two-year contracts.

⚠ Contract type is the single strongest predictor of churn in this dataset.

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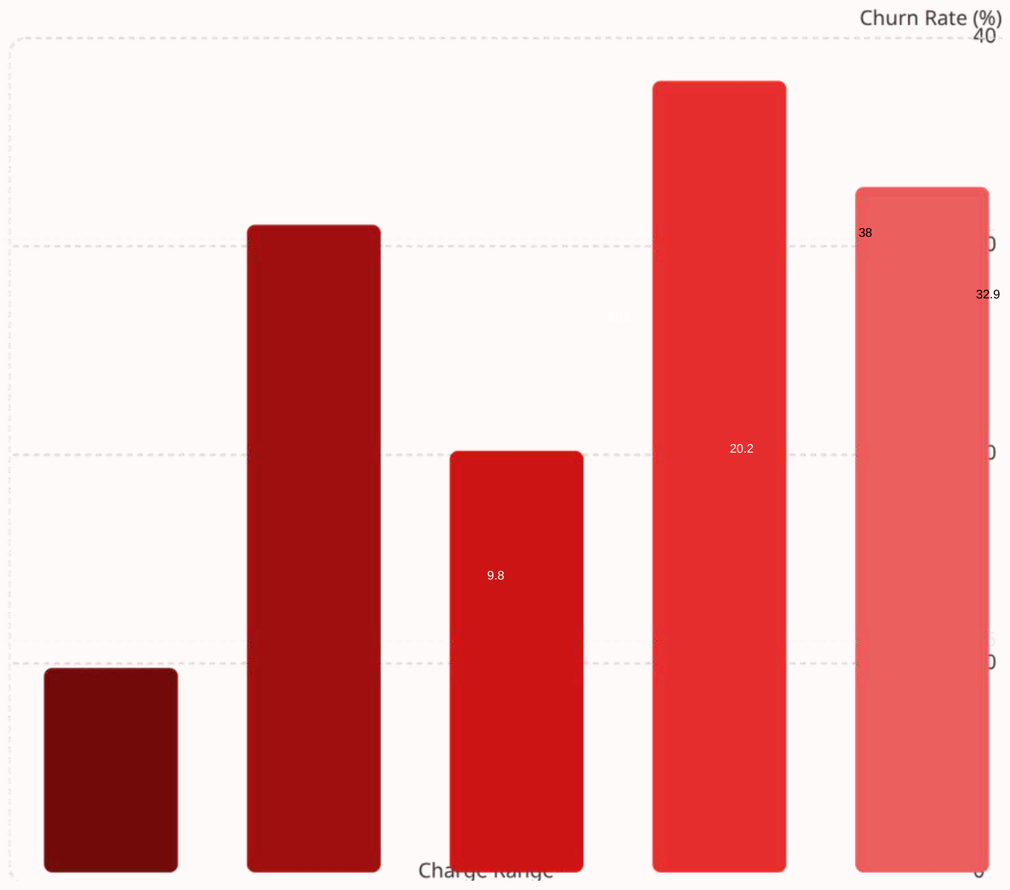
Churn by Monthly Charges & Internet Service



Internet Service Churn Rates

Service Type	Churn Rate
Fiber Optic	41.89%
DSL	18.96%
No Internet	7.40%

Churned customers pay on average **$74.44/mo** vs. **$61.27/mo** for retained customers.

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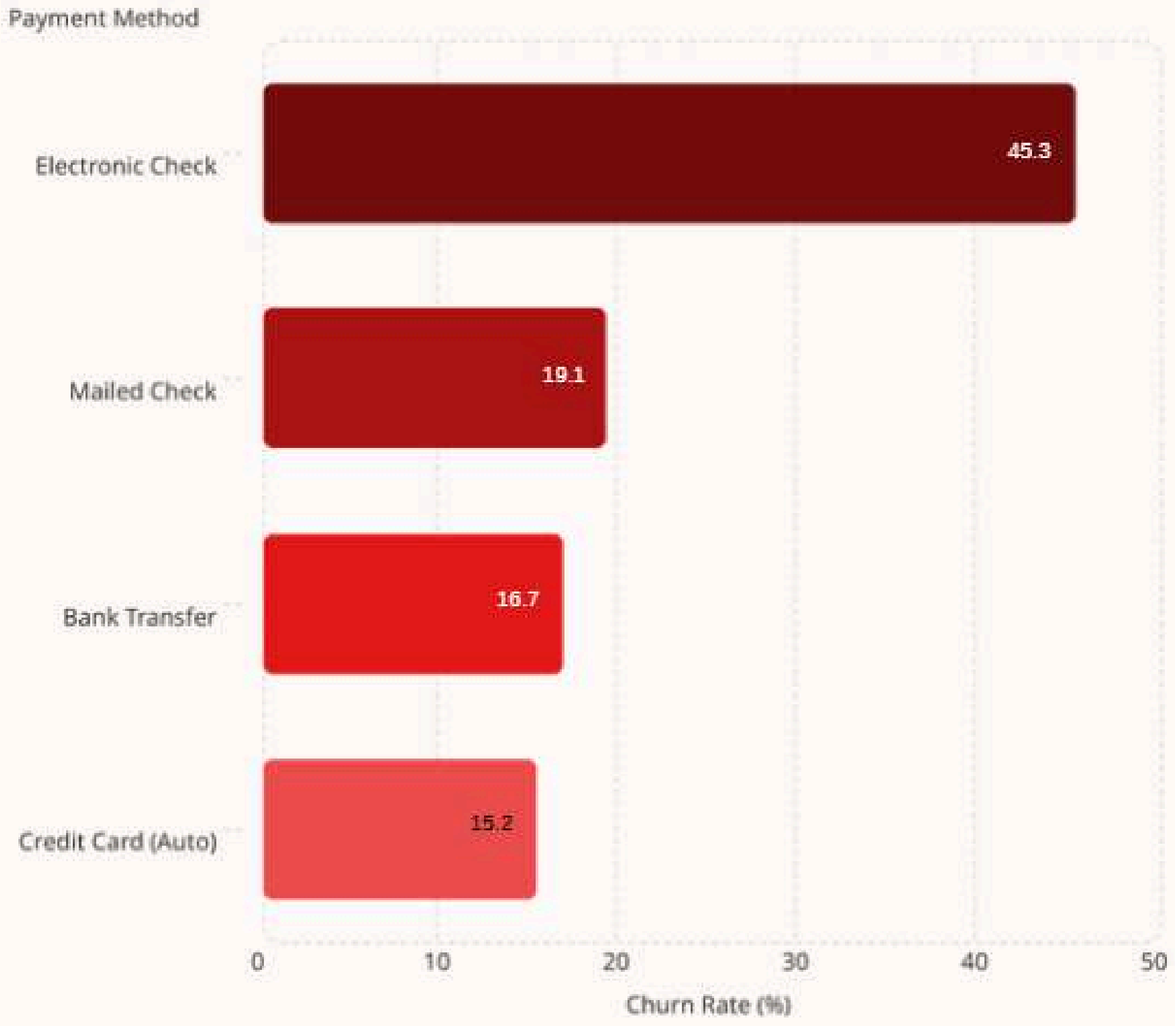
Support Services & Payment Method Impact

Churn Rate by Support Coverage

Online Security	Device Protection	Churn Rate
No	No	47.05%
No	Yes	32.95%
Yes	No	20.04%
Yes	Yes	10.17%

Tech Support alone drops churn from **41.64%** to **15.17%**.

Payment Method Churn Rates



Key Insights

Contract Type

Month-to-month customers churn at **42.71%** vs. 2.83% for two-year contracts.

Monthly Charges

Higher charges correlate with higher churn — churned customers pay ~$13 more per month.

Tenure

New customers churn most. Long-term customers show strong loyalty.

Support Services

Customers without Tech Support, Security, or Device Protection churn significantly more.

Payment Method

Electronic check users churn at **45.29%** vs. ~15% for automatic payment users.

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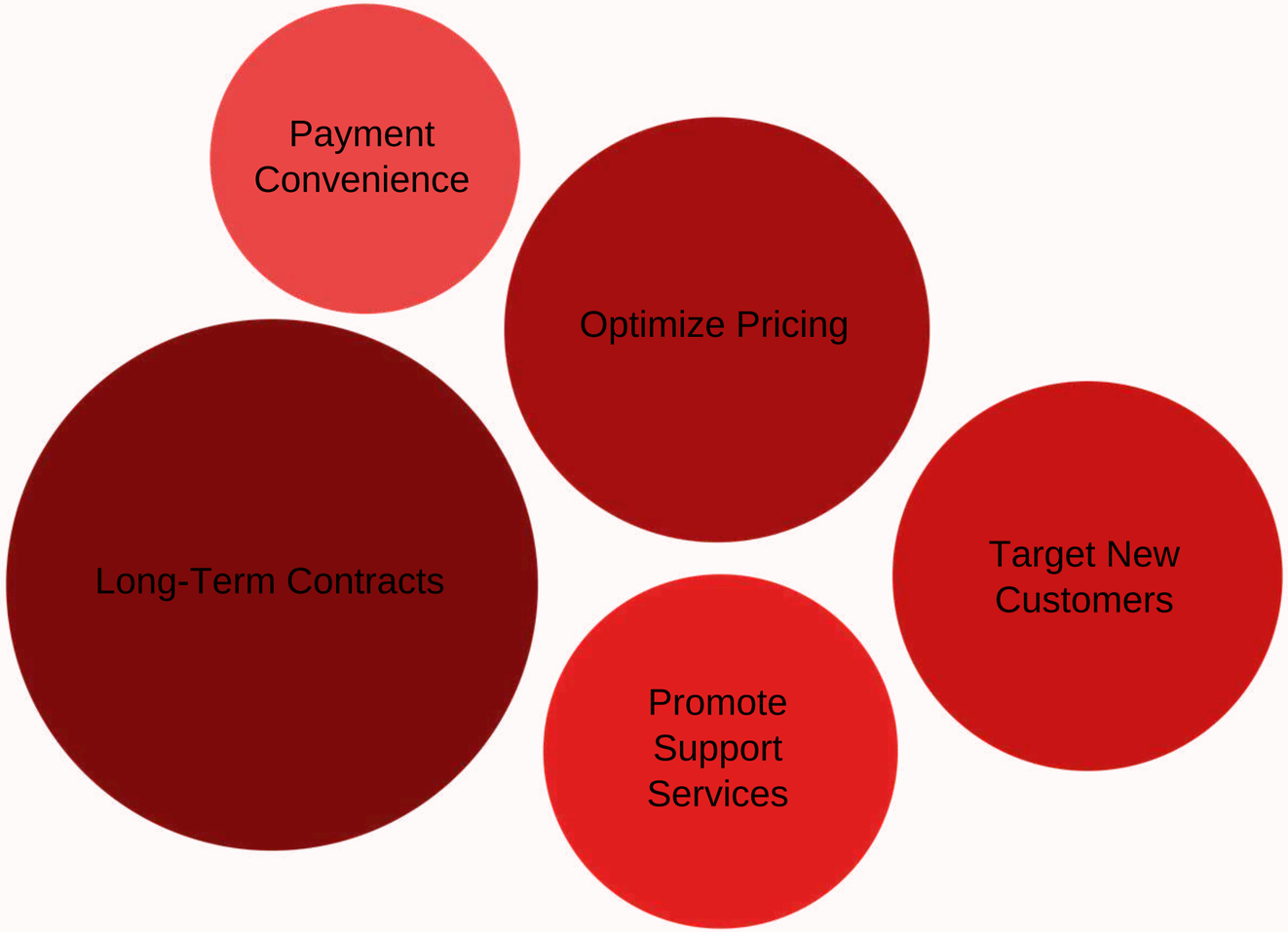
Retention Strategy Recommendations



These five data-driven actions directly address the top churn drivers identified in the analysis.

Strategy Deep Dive

Encourage Long-Term Contracts

Promote yearly and two-year plans with incentives to reduce month-to-month churn.

Optimize Pricing

Review high monthly charges and bundle value-added benefits to justify costs.

Focus on New Customers

Improve early onboarding experience — churn risk is highest in the first months.

Promote Support Services

Increase awareness of Tech Support, Online Security, and Device Protection.

Improve Payment Convenience

Nudge customers toward automatic payment methods to reduce churn risk.

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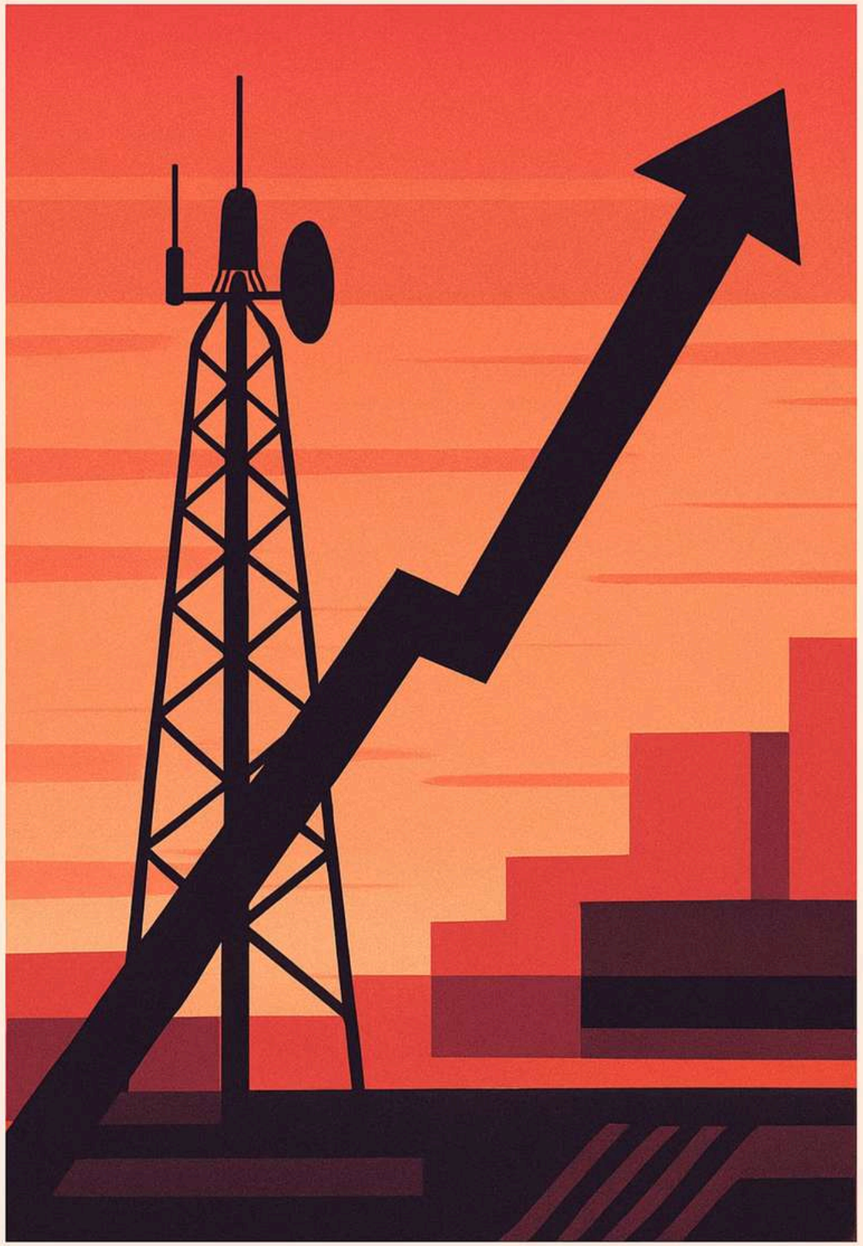

From Churn to Loyalty

By acting on contract structure, pricing, onboarding, support services, and payment methods, this telecom company can meaningfully reduce churn and build a more loyal customer base.

Tools Used

Python · SQL (PostgreSQL) · Power BI

Dataset

7,043 customers · 21 features · Churn labeled

Overall Churn Rate

26.5% of customers churned

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